

Apples and Quantum Mechanics

John Regis

12/04/2026

Chapter 1

Understanding Time-Dependent Perturbation Theory Using Apples

Why apples?

I was studying time-dependent perturbation theory and I was wondering how it actually works. At some point I remembered my elementary school math teacher saying: if you don't understand math, think in terms of apples. Back then it was about fractions. Now unfortunately it is about quantum mechanics.

So I decided: fine, let's do perturbation theory using apples.

The system

Assume we have exactly one apple. God gives this apple to Calvin and says: don't eat it. Since it is Calvin, he eats it anyway (at least that is the concern).

We define two states:

$|o\rangle$ = apple outside (not eaten)

$|i\rangle$ = apple inside Calvin (eaten)

Now, instead of saying the apple is definitely in one place, we allow the system to be in a combination of both:

$$|\Psi\rangle = c_o|o\rangle + c_i|i\rangle$$

Here, c_o and c_i are complex numbers called amplitudes.

Now comes the important part.

$|c_o|^2$ = probability that the apple is outside

$|c_i|^2$ = probability that the apple is inside Calvin

So the squares of these numbers tell us what we would actually observe if we checked where the apple is.

Since there is only one apple, the probabilities must add up to 1:

$$|c_o|^2 + |c_i|^2 = 1$$

This is just conservation of apple.

Now, a slightly uncomfortable but important point:

Before we measure, the apple is not strictly outside or inside. The system is in a superposition, meaning it has some probability of being found in either state.

So in this description:

- c_o tells us “how much the system leans toward the apple being outside”
- c_i tells us “how much the system leans toward the apple being inside”

What is the Hamiltonian anyway?

So, the Hamiltonian is basically a protein-checking machine.

You give it a state, and it tells you how much “stuff” (energy, protein, whatever we decided today) is in that state. In our case, we are emotionally invested in protein.

So when it acts on the apple:

$$\hat{H}_0|o\rangle = E_o|o\rangle$$

it is essentially saying: “this apple has E_o grams of protein.”

Same for the eaten apple:

$$\hat{H}_0|i\rangle = E_i|i\rangle$$

So yes, mathematically it is energy, but since everyone is obsessed with protein nowadays, we are calling it protein. You can reinterpret it as actual energy if you want, for example, the kinetic energy of the apple when someone throws it at Calvin for his horrendous rubbishness(kona). That is also a valid physical picture.

But for now, we keep things civil and stick to protein.

The important point is: the Hamiltonian does not change the state here, it just reads off a number associated with it. It is like a measurement device that says:

“This state has this much energy (or protein).”

After getting an expert opinion from Mr. Anandu, who is an expert in, well, I don’t know what he does, nor does anyone, including him, but always talks about some fields which I never understood, I want to acknowledge the fact that the statement “the Hamiltonian does not change the state here” is misleading and should be revised.

That is true; here, the time evolution operator changes the state’s phase:

$$e^{-iE_nt/\hbar}$$

This exponential term changes constantly with time, even when there is no perturbation because of the Hamiltonian H_0 .

A proper correction would be:

“The H_0 does not change the probabilities or the transition amplitude.”

Time enters the story

Now the story continues.

So this apple, after 1 second Calvin will take a bite and 1/10 of the apple will go into his stomach. After 2 seconds, 2/10 of the apple is gone. And so on. After 10 seconds, the apple will be completely inside Calvin.

This is the classical picture of how things happen. The apple is gradually disappearing in a very predictable and slightly unfortunate way.

Now, in quantum mechanics, we cannot directly say “1/10 of the apple is gone”. That would be too nice and too sensible. Instead, we describe everything in terms of probabilities.

So rather than tracking how much of the apple is gone, we track how likely it is to find the apple outside or inside.

To bring time into the picture, we introduce the time evolution form. If you are somehow familiar with the factor $e^{-iE_nt/\hbar}$, we can write the general state as:

$$|\Psi(t)\rangle = \sum_n c_n(t) e^{-iE_nt/\hbar} |n\rangle$$

Since we only have two states ($|o\rangle$ and $|i\rangle$), this becomes:

$$|\Psi(t)\rangle = c_o(t) e^{-iE_ot/\hbar} |o\rangle + c_i(t) e^{-iE_it/\hbar} |i\rangle$$

Now here is how the classical and quantum pictures connect (very loosely):

- “1/10 of the apple is gone” $\rightarrow$ small increase in $|c_i(t)|^2$
- “2/10 of the apple is gone” $\rightarrow$ larger increase in $|c_i(t)|^2$
- ...
- “apple completely gone” $\rightarrow |c_i(t)|^2 \approx 1$

So the idea is not that the apple is literally splitting into fractions, but that the *probability* of finding it inside Calvin increases with time.

Meanwhile, the exponential factors $e^{-iE_nt/\hbar}$ are just there, quietly doing their job, rotating phases in the background like responsible adults, while Calvin causes all the visible chaos through the change in $c_o(t)$ and $c_i(t)$.

Putting everything into the Schrödinger equation

Since we now have these nice formulas for our Calvin scenario, we should actually use them and see what happens.

The starting point is the Schrödinger equation:

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = (\hat{H}_0 + \hat{V}) |\Psi(t)\rangle$$

Now we already know the form of the state:

$$|\Psi(t)\rangle = \sum_n c_n(t) e^{-iE_nt/\hbar} |n\rangle$$

So we simply substitute this into the equation and see what breaks

Left-hand side

First, take the time derivative:

$$\frac{d}{dt} |\Psi(t)\rangle = \sum_n \left(\dot{c}_n(t) e^{-iE_nt/\hbar} - \frac{iE_n}{\hbar} c_n(t) e^{-iE_nt/\hbar} \right) |n\rangle$$

Multiplying by $i\hbar$:

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \sum_n (i\hbar \dot{c}_n(t) - E_n c_n(t)) e^{-iE_nt/\hbar} |n\rangle$$

Right-hand side

Now apply $(\hat{H}_0 + \hat{V})$:

First the easy part:

$$\hat{H}_0|\Psi(t)\rangle = \sum_n E_n c_n(t) e^{-iE_n t/\hbar} |n\rangle$$

Now the slightly more dangerous part:

$$\hat{V}|\Psi(t)\rangle = \sum_n c_n(t) e^{-iE_n t/\hbar} \hat{V}|n\rangle$$

So the full right-hand side becomes:

$$(\hat{H}_0 + \hat{V})|\Psi(t)\rangle = \sum_n E_n c_n(t) e^{-iE_n t/\hbar} |n\rangle + \sum_n c_n(t) e^{-iE_n t/\hbar} \hat{V}|n\rangle$$

Now compare both sides

On the left we had:

$$\sum_n (i\hbar \dot{c}_n - E_n c_n) e^{-iE_n t/\hbar} |n\rangle$$

On the right we have:

$$\sum_n E_n c_n e^{-iE_n t/\hbar} |n\rangle + \sum_n c_n e^{-iE_n t/\hbar} \hat{V}|n\rangle$$

Now something nice happens:

The $E_n c_n$ terms cancel on both sides.

So we are left with:

$$\sum_n i\hbar \dot{c}_n(t) e^{-iE_n t/\hbar} |n\rangle = \sum_n c_n(t) e^{-iE_n t/\hbar} \hat{V}|n\rangle$$

What does this mean?

This is the key point:

The change in the coefficients $c_n(t)$ is entirely driven by the perturbation $\hat{V}$.

In our story:

- $\hat{H}_0$ just sits there measuring protein like a responsible machine
- $\hat{V}$ (Calvin) is the one actually changing where the apple is

So all the “apple eating dynamics” comes from the perturbation.

Next step (important)

To go further, we would project onto a specific state (like $|o\rangle$ or $|i\rangle$), and that will give us equations for $c_o(t)$ and $c_i(t)$.

That is where the actual dynamics (how fast Calvin eats, and how quantum mechanics allows him to uneat it) shows up.

Why projection actually works

We still need to figure out what these c_o and c_i actually are. Right now they are just mysterious coefficients doing all the work behind the scenes.

To extract them, we use one simple trick: **projection onto states**.

Take the inner product:

$$\langle i|o\rangle$$

What does this mean?

It is basically asking:

“How much of the apple that is outside is also inside Calvin’s stomach?”

Well... zero. Unless Calvin is an alien, which, to be fair, is not completely ruled out.

So:

$$\langle i|o\rangle = 0$$

Now take:

$$\langle i|i\rangle$$

This is like asking:

“How much of the apple that is inside Calvin is actually inside Calvin?”

The answer is obviously:

$$1$$

No shit Sherlock.

Similarly:

$$\langle o|o\rangle = 1, \quad \langle o|i\rangle = 0$$

So what do we learn from this?

These states are **orthonormal**:

- orthogonal: different states give 0
- normalized: same states give 1

Now here is the trick.

Start from:

$$|\Psi\rangle = c_o|o\rangle + c_i|i\rangle$$

Project onto $\langle o|$:

$$\langle o|\Psi\rangle = c_o\langle o|o\rangle + c_i\langle o|i\rangle$$

$$= c_o(1) + c_i(0) = c_o$$

Boom. We got:

$$c_o = \langle o|\Psi\rangle$$

Now do the same for $\langle i|$:

$$\langle i|\Psi\rangle = c_o\langle i|o\rangle + c_i\langle i|i\rangle$$

$$= 0 + c_i(1) = c_i$$

So:

$$c_i = \langle i|\Psi\rangle$$

What just happened?

Projection literally picks out the coefficient.

That's it.

All this formalism just to say:

“If you want to know how much of the system is in a state, project onto that state.”

In apple language:

- $\langle o | \Psi \rangle$ = how much the apple is outside
- $\langle i | \Psi \rangle$ = how much the apple is inside Calvin

Projection and final expression for $c_m(t)$

Starting from:

$$\sum_n i\hbar \dot{c}_n(t) e^{-iE_n t/\hbar} |n\rangle = \sum_n c_n(t) e^{-iE_n t/\hbar} \hat{V} |n\rangle$$

Project onto $\langle m |$:

$$i\hbar \dot{c}_m(t) e^{-iE_m t/\hbar} = \sum_n c_n(t) e^{-iE_n t/\hbar} \langle m | \hat{V} | n \rangle$$

Define the matrix elements:

$$V_{mn} = \langle m | \hat{V} | n \rangle$$

So:

$$i\hbar \dot{c}_m(t) e^{-iE_m t/\hbar} = \sum_n c_n(t) e^{-iE_n t/\hbar} V_{mn}$$

Divide both sides by $e^{-iE_m t/\hbar}$:

$$i\hbar \frac{dc_m(t)}{dt} = \sum_n c_n(t) V_{mn} e^{i(E_m - E_n)t/\hbar}$$

This gives the general expression for how the coefficients evolve in time under the perturbation.

Interpretation

Each coefficient $c_m(t)$ changes due to:

- coupling to other states through V_{mn}
- phase factors $e^{i(E_m - E_n)t/\hbar}$

So in the apple picture:

- the coefficients change because Calvin (the perturbation) mixes states
- the exponential factors track how different energy states interfere over time

Solving the two-state system (the actual payoff)

We start from the state:

$$|\Psi(t)\rangle = c_o(t)e^{-iE_o t/\hbar}|o\rangle + c_i(t)e^{-iE_i t/\hbar}|i\rangle$$

Evolution equation

The general result is:

$$i\hbar \frac{dc_m}{dt} = \sum_n c_n(t) V_{mn} e^{i(E_m - E_n)t/\hbar}$$

Matrix elements

We choose a simple coupling:

$$V_{io} = g, \quad V_{oi} = g$$

All other matrix elements are zero.

Coupled equations

For $c_o(t)$:

$$i\hbar \frac{dc_o}{dt} = g c_i(t) e^{i(E_o - E_i)t/\hbar}$$

For $c_i(t)$:

$$i\hbar \frac{dc_i}{dt} = g c_o(t) e^{i(E_i - E_o)t/\hbar}$$

Degenerate case

Now take the special case:

$$E_o = E_i$$

Then:

$$e^{i(E_o - E_i)t/\hbar} = 1$$

So the equations simplify to:

$$\begin{aligned} i\hbar \frac{dc_o}{dt} &= g c_i(t) \\ i\hbar \frac{dc_i}{dt} &= g c_o(t) \end{aligned}$$

Differentiate once more:

$$\frac{d^2 c_o}{dt^2} = -\frac{g^2}{\hbar^2} c_o$$

This gives:

$$c_o(t) = \cos\left(\frac{gt}{\hbar}\right)$$

Using the first equation:

$$c_i(t) = -i \sin\left(\frac{gt}{\hbar}\right)$$

Final state

$$|\Psi(t)\rangle = \cos\left(\frac{gt}{\hbar}\right) e^{-iEt/\hbar} |o\rangle - i \sin\left(\frac{gt}{\hbar}\right) e^{-iEt/\hbar} |i\rangle$$

Probabilities

$$|c_o(t)|^2 = \cos^2\left(\frac{gt}{\hbar}\right)$$

$$|c_i(t)|^2 = \sin^2\left(\frac{gt}{\hbar}\right)$$

Interpretation

The interpretation should occur naturally since that was the whole point of this silly apple physics. The particular form we got for the probability, which is basically the square of the amplitude, shows us the obvious. The cosine function, which starts from 1, signifies that there was a whole apple there, and it slowly gets devoured by hungry Calvin, who usually eats 20 chapatis for a day; also, the coefficient that signifies how much of the apple is in Calvin's stomach starts to grow. An important fact to note here is that we did not set any bounds on the time, and therefore, after some time, the apple starts to appear outside Calvin's stomach, which is something I really don't want to discuss or imagine. But you should always be mindful of the fact that the probabilities oscillate.

Conclusion

The whole note is not a study material for a formal study, but rather a silly note or a comment on the topic of Time-dependent perturbation. That might help you to understand what the terms that come along with the mathematical derivation, especially a small intuition on how the coefficients are working here, can be achieved from this, I guess.

Chapter 2

Bardeen's Theory of Tunnelling and Apples

Arjun with his new job

Arjun, a friend of mine, unlike Calvin, is a man of purpose. He is not interested in gobbling apples like Calvin; rather, he has a job. The job is very simple: moving apples from one shelf to another. He cannot do it randomly, of course he has certain limitations that are attributed to the type of shelf he deals with and also to his own nature.

He, like particles in quantum mechanics, does everything probabilistically. Will he eat food for his breakfast? I do not know. If he is starving, yes. If he is not that hungry, the chances are obviously low. Similarly, will he do his job of picking up apples from the shelf? It depends... depends on certain factors.

The new job he has is very unique and very basic. He will be given two racks that have 100 shelves, partially filled with apples. He has to move apples from the left side rack to the right side rack (he cannot go otherwise).

For now, let us scale down the 100 apples to 1. That is, 0.5 apple means 50 apples, and 0.12 apples means 12 apples (I will explain later why I chose this scaling).

Now, the peculiar probabilistic nature of Arjun makes him do something slightly irrational. Let us say the left-side rack comes with 100 apples, and the right-side rack has zero apples. That is, the number of apples is 1 and 0 (remember our scaling). In that case, he will certainly transfer the apple with probability 1.

Likewise, if the left-side rack is completely empty (zero apples), and the right-side rack already has 100 apples (that is, 1 unit of apple), then he cannot transfer any apple. The probability of transferring an apple is zero.

The System

As we always do, let us define some states. What are the states here? Unlike last time, we cannot model this with just two states.

We can immediately observe that there are two sets of states:

$$|\psi^R\rangle \quad \text{and} \quad |\psi^L\rangle,$$

which denote the right side and the left side.

Now, in order to represent the many energy levels (say, 100 levels) in each side, we introduce quantum numbers μ and ν . Thus, the states are written as:

$$|\psi_\mu^L\rangle \quad \text{and} \quad |\psi_\nu^R\rangle.$$

To incorporate the probabilistic nature, we will try a simple model. Let the number of apples arranged in the left shelves be denoted by $f(E_L)$, and the number of apples arranged in the right shelves be denoted by $f(E_R)$.

Now consider the following product:

$$P = f(E_L) [1 - f(E_R)].$$

Let P represent the probability of transition.

If the left side has 100 apples (i.e., $f(E_L) = 1$) and the right side has zero apples (i.e., $f(E_R) = 0$), then the product gives a transition probability of 1.

Similarly, if the right side does not have any available space in its shelves (i.e., $f(E_R) = 1$), then the probability drops to zero.

This model may have certain limitations, but for now we will keep this form of probability and use it to build intuition toward Bardeen's theory of tunneling and time-dependent perturbation theory.